

CS471 – Lesson 4 In-Class Worksheet

Search Problems & Problem Formulation

Guided In-Class Worksheet (Instructor-Led)

Objective: Walk through the formal definition of a search problem using a standard grid environment.

Scenario: A reconnaissance UAV must navigate a hostile sector represented by an $M \times N$ grid. The objective is to find a safe flight path from its launch point at $(0, 0)$ in the bottom left corner to the extraction point at (M, N) in the top right corner.

Problem Formulation

Describe the problem using the formal terms of a search problem:

1. **States:**

2. **Actions (and costs):**

3. **Successor Function:**

4. **Start State:**

5. **Goal Test:**

State Space Evaluation (same scenario)

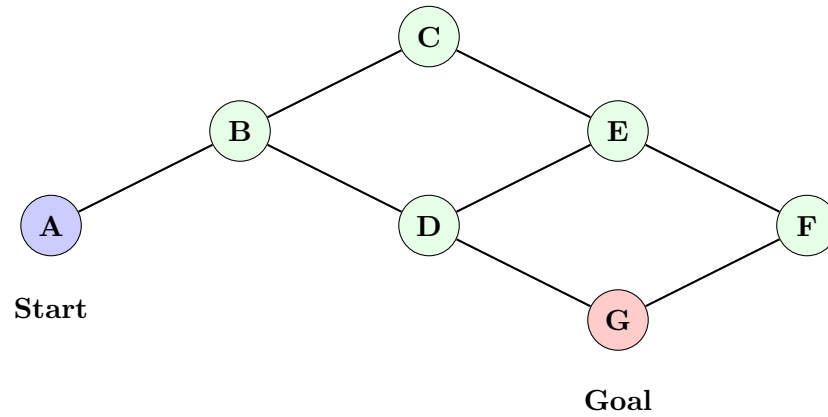
6. **What is the size of the State Space for this problem?**

7. Draw the first three layers of the Search Tree (Assume $M = 3, N = 3$):

Student Practice

Objective: Independently formulate a search problem based on a non-grid waypoint map.

Scenario: Now consider a path problem similar to a map. A ground convoy must navigate a secure road network. The map consists of specific tactical waypoints (A through G) connected by roads.



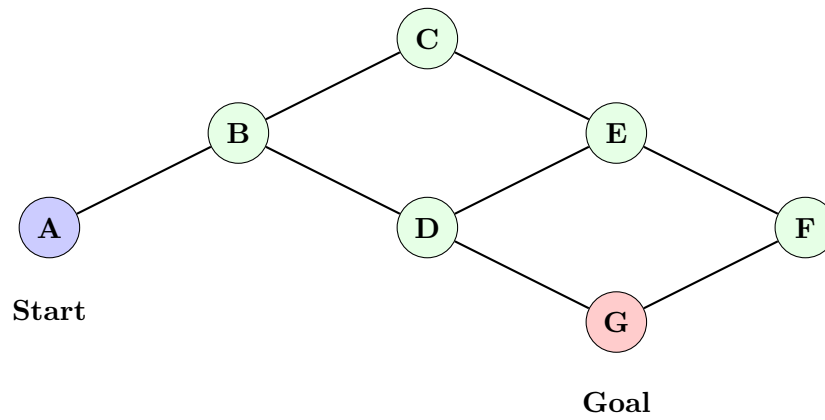
Problem Formulation

For the map, describe the core components of the search problem:

1. **States:**
2. **Actions (and costs):**
3. **Successor Function:**
4. **Start State / Goal Test:**

Search Space Evaluation

5. Draw the Search Tree:



(Same as above) - Trace the potential paths from Node A up to a depth of 3. Note any redundant paths (cycles).